

C1 implements I1

Interface Implementation in Java

1. Aim

To demonstrate how a class implements an interface using the "implements" keyword in Java.

2. Definition

An interface is a blueprint that defines methods which can be implemented by a class.

A class uses the "implements" keyword to provide implementations for the methods declared in an interface.

Syntax:

```
class C1 implements I1
{
    // implementation of interface methods
}
```

"I1 implements C1" is invalid because an interface cannot implement a class."

3. Structure

```
    I1
    Interface
    ↑
    implements
    |
    C1
    Class
```

4. Suitable Example

Here, "I1" represents a Printable interface, and "C1" represents a Student class that implements it.

5. Program

```
interface I1
{
    void printDetails();
}

public class C1 implements I1
{
    public void printDetails()
    {
        System.out.println("Student details printed successfully");
    }

    public static void main(String[] args)
    {
        C1 obj = new C1();

        // Test Case 1
        System.out.println("Test Case 1:");
    }
}
```

```
        obj.printDetails();

        // Test Case 2
        System.out.println("\nTest Case 2:");
        obj.printDetails();

        // Test Case 3
        System.out.println("\nTest Case 3:");
        obj.printDetails();
    }
}
```

6. Sample Output

```
Test Case 1:
Student details printed successfully

Test Case 2:
Student details printed successfully

Test Case 3:
Student details printed successfully
```

7. Test Case Verification

Test Case	Method Called	Expected Output	Result
TC1	printDetails()	Student details printed successfully	Passed
TC2	printDetails()	Student details printed successfully	Passed
TC3	printDetails()	Student details printed successfully	Passed

8. Conclusion

Thus, the program successfully demonstrates interface implementation in Java using "C1 implements I1". The interface method was successfully implemented by class "C1", and all test cases were verified successfully.